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Motivation

- **Global Context:** In an era of increasing climatic volatility, the demand for precise, localized weather information has never been higher.
- **The Problem:** Contemporary forecasting is dominated by resource-heavy Numerical Weather Prediction (NWP) models or opaque consumer apps that lack transparency.
- **Objective:** Developing a lightweight, computationally efficient solution that captures localized data and performs frequency/trend analysis for explainable results.

Theoretical Foundation

The study addresses distinct temporal behaviors based on two scientific pillars[cite: 99]:

- **Persistence and Momentum Theory:** Evolving the standard persistence method by treating the atmosphere as a fluid system with thermal inertia.
- **Stochastic Volatility:** Addressing variables like relative humidity which are sensitive to immediate localized fluctuations.
- **Memory Decay:** Prioritizing recent observations while mathematically "forgetting" distant history.

The Mathematical Core

Predictive capability is categorized into components optimized for specific physical characteristics.

- **Temperature Trends (SLR):** Modeled using Linear Least Squares to minimize the Residual Sum of Squares (RSS).
- **Formula:** $m = \frac{n \sum (x_i y_i) - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$.
- **Humidity Dynamics (EWMA):** Implementing a recursive weighted average to react quickly to incoming fronts.
- **Formula:** $S_t = \alpha \cdot Y_t + (1 - \alpha) \cdot S_{t-1}$.

System Architecture

A modular full-stack architecture ensuring heavy mathematical lifting remains server-side[cite: 108].

- **Backend:** Developed with Flask to manage request orchestration and JSON payloads.
- **Frontend:** An asynchronous Single Page Application (SPA) using a card-based layout and dynamic CSS3 gradients.
- **Data Orchestration:** Utilizing the Open-Meteo API for 10-day historical training sets.

Technical Implementation

- **Physical Clamping:** A constraint layer ensures outputs remain within physical boundaries (e.g., [0, 100] for humidity).
- **Heuristic Thresholding:** A 0.3mm floor filters out meteorological noise to prevent false precipitation predictions.
- **Validation Framework:** A "Trust Score" where every $1^{\circ}C$ of error reduces the AI's accuracy rating by 10%.

Challenges & Solutions

- **Persistence Bias:** The model assumes the future is a continuation of a trend and can miss non-linear turning points.
- **Portability Solution:** Algorithms rely on simple arithmetic, allowing deployment on low-power 8-bit or 16-bit microcomputers (Arduino).
- **Edge Intelligence:** Standalone devices can perform local forecasting autonomously without internet connectivity.

Conclusion & Future Outlook

- **Summary:** The engine provides a verifiable, low-latency alternative to traditional global forecasting models.
- **Future Work:** Incorporating barometric pressure trends to detect non-linear atmospheric shifts.
- **Applications:** Expanding into precision agriculture, smart HVAC management, and renewable energy forecasting.

Thank you for your attention!

Advancing Localized and Explainable AI Analytics.

Links & References

- **Open-Meteo API:** https://open-meteo.com/
- **Linear Regression:** Montgomery, D. C. et al. (2021)
- **EWMA Foundation:** Hunter, J. S. (1986). “The Exponentially Weighted Moving Average”